



Department of Computer Science and Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch : IV Semester B.E CSE
Course Code & Title : CS8492 & Database Management Systems
Name of the Faculty member : Mr. K.Vignesh Saravanan, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit III / Transactions / Deadlock – Prevention & Avoidance/Detection
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO 8
- **Activity Chosen:** Flipped Classroom
- **Justification:**

Flipping the classroom is an inverting the classroom approach to teaching. In this approach, the traditional in-class teaching is “flipped” to better meet the needs of individual learners. Students gain control of the learning process through studying course material outside of class, using readings, pre-recorded video lectures. It helps the faculty/lecturer to redefine in-class activities and include homework problems and keep the students engaged in the content.

- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
 - **Plan:** Basics of Deadlock is already learnt in the subject operating systems and few topics related to DBMS is given as self-learning through self-exploration.
 - **Identify and Share:** Related materials are identified and I posted relevant references/materials/notes of deadlock mechanisms to the students and given as self-learning.
 - **Evaluate:**
 - I have prepared few questions related to the content shared and ask those questions in next class session and make the students to write/present in class.
 - Make the students to find out the answer by their own by learning.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO7	PO8	PO9	PO10	PO12	PSO1
CO1	3	3	3	1	1	1	1	2	3
(1 – Low			2 – Moderate			3 – High)			

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PO12
	2	2	3	3	3	2
Justification for correlation	Apply basic Knowledge and fundamentals in Transactions and deadlocks	Identify the need of deadlock prevention and detection	Able to design and develop the mechanism for the deadlock prevention & avoidance	Functional individually in identifying the protocols for deadlock prevention and avoidance	Communicate / share the ideas with other students in written and oral presentation	Ability to reproduce the contents gathered through self-learning

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Student felt good, since they can study at their own pace/time.
- They felt that through such learning, they can explore more.

❖ ***Benefit of the practice:***

- More one-to-one time between teacher and student.
- More collaboration time for students.
- Students learn at their own pace.
- Practical things – like missing class due to illness – become less problematic.
- It encourages students to come to class prepared.

❖ ***Challenges faced in implementation:***

- Relies on student preparation – few students did not come prepared.
- The depth of the subject can be dictated by the student themselves or the group the student is working with.
- The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- ❖ <https://www.teachthought.com/learning/the-definition-of-the-flipped-classroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20independently%20at%20home.>
- ❖ https://en.wikipedia.org/wiki/Flipped_classroom
- ❖ [youtube.com/watch?v=BCIxikOq73Q](https://www.youtube.com/watch?v=BCIxikOq73Q)
- ❖ <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>

Signature of Faculty Member

HOD